

REDUCTIVE CONVERSION OF NITRO ALKENES TO KETONES AND/OR OXIMES
IN AN AQUEOUS HClO_4 - CH_2Cl_2 -DIOXANE-(Pb) SYSTEM

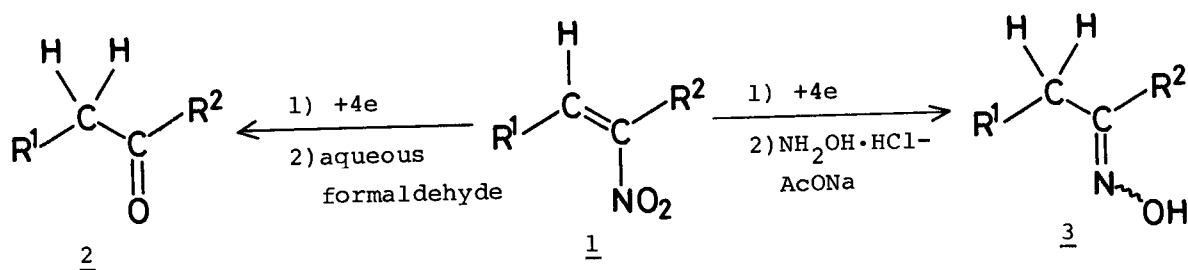
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Electrochemical and chemical reduction of nitro alkenes in an aqueous HClO_4 - CH_2Cl_2 -dioxane-(Pb) system afforded ketones and oximes in good yields, each of which can be obtained selectively by treating with either aqueous formaldehyde or hydroxylamine as a proper workup process, respectively.

Nitro alkenes 1, readily prepared by the condensation of aldehydes with nitro alkanes,¹⁾ are synthetically equivalent to the corresponding ketones 2 and oximes 3. Namely, nitro alkenes 1 can be converted to 2 and/or 3 by the reduction with Fe-dilute HCl,^{2a)} Zn-AcOH,^{2b)} TiCl_3 - NH_4OAc -aqueous methanol,^{2c)} and others.³⁾ Electrolytic reduction of 1 to 2 and/or 3 has been also investigated by several groups.⁴⁾ However, the product selectivity and the yield for the desired products 2 and 3 are not satisfactory in a practical sense.⁵⁾ In this communication, we describe some of potential procedures for the reductive conversion of 1 to 2 or 3 assisted with lead in an aqueous HClO_4 - CH_2Cl_2 -dioxane system, which involve electrochemical and chemical methods followed by a proper workup process.



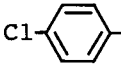
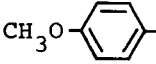
The electrochemical reduction was carried out in an undivided cell fitted with two lead electrodes ($1.5 \times 2 \text{ cm}^2$). A typical electrolysis procedure is as follows: To a solution of nitro alkene 1a ($R^1 = p\text{-chlorophenyl}$; $R^2 = \text{CH}_3$, 0.53 mmol) in dichloromethane (2 ml) and dioxane (0.5 ml) was added aqueous 20% perchloric acid (10 ml) and the two-phase solution was electrolyzed under vigorous stirring at 10 mA/cm^2 , changing the current direction every 30 s by a commutator, at room temperature. After passage of 3 F/mol of electricity, aqueous 37% formaldehyde (1 ml) was added and the stirring was continued for additional 20 min without passing current. The usual workup of the organic phase afforded the ketone 2a ($R^1 = p\text{-chlorophenyl}$; $R^2 = \text{CH}_3$) in 93% yield (Procedure A: entry 1 in the Table).

On the other hand, after the electrolysis (3.5 F/mol of electricity passed), the organic phase was separated from the aqueous phase and treated with $\text{NH}_2\text{OH}\cdot\text{HCl}$ (300 mg) and sodium acetate (370 mg) at room temperature for 2 h, affording the corresponding oxime 3a ($R^1 = p\text{-chlorophenyl}$; $R^2 = \text{CH}_3$) in 90% yield (Procedure B: entry 2 in the Table). Apparently, the initial product of the reduction was a mixture of 2 and 3 and the suitable workup of the products allowed to isolate either 2 or 3 respectively.⁶⁾

Actually, the transformation of 1 into 2 or 3, which is a four-electron reduction process,⁷⁾ consumed 2-3.5 F/mol of electricity and after the electrolysis (entry 1 in the Table) lead(II) salts were obtained from the aqueous layer.⁸⁾ This suggests that the reduction of 1 to 2 or 3 (Procedure A and B) involves both electrochemical reduction and chemical reduction with electrochemically activated lead(0). In fact, the reduction of 1 with lead in the same media occurred smoothly without passing current.⁹⁾ Thus, prior to the reduction the current (10 mA/cm^2) was passed between two lead electrodes in aqueous 20% perchloric acid for 10 min in order to activate the metal surface. To this solution a dichloromethane solution of 1a was added and the mixture was stirred for 30 min at room temperature. Workup in accordance with that of Procedure A afforded 2a in 90% yield (Procedure C: entry 3 in the Table). The treatment of 1a in the same media with lead¹⁰⁾ without activating its surface resulted in the recovery of most of 1a along with a small amount of 2a ($\sim 10\%$) after stirring for 1 h.

Electrochemical reduction of 1 proceeded mainly when the electrolysis was

Table Reduction of Nitro Alkenes in an Aqueous HClO_4 - CH_2Cl_2 -dioxane-(Pb) System

entry	nitro alkene		procedure ^{a)}	electricity F/mol	product, yeild % ^{b)}	
	R ¹	R ²			ketone <u>2</u>	oxime <u>3</u>
1		CH ₃ -	<u>1a</u> A	3	93	--
2			<u>1a</u> B	3.5	--	91
3			<u>1a</u> C	--	91	--
4			<u>1a</u> D	4.3	83	--
5		CH ₃ -	<u>1b</u> A	3	80	--
6			<u>1b</u> B	3	--	75
7			<u>1b</u> C	--	76	--
8			<u>1b</u> D	4	73	--
9		CH ₃ -	<u>1c</u> A	3	87	--
10			<u>1c</u> B	3	--	88
11			<u>1c</u> C	--	80	--
12			<u>1c</u> D	4.4	74	--
13		CH ₃ -	<u>1d</u> A	2	70	--
14			<u>1d</u> B	3	--	55
15			<u>1d</u> C	--	71	--
16			<u>1d</u> D	4.1	63	--
17	C ₆ H ₁₃ -	CH ₃ -	<u>1e</u> A	2.7	81	--
18			<u>1e</u> B	3	--	67
19			<u>1e</u> C	--	88	--
20			<u>1e</u> D	4	80	--
21	C ₆ H ₁₃ -	CH ₃ CO(CH ₂) ₂ -	<u>1f</u> A	3	95	--
22			<u>1f</u> D	4	75	--

a) Procedure A: worked up with aqueous 37% formaldehyde; B: worked up with $\text{NH}_2\text{OH}\cdot\text{HCl}$ -NaOAc; C: reduction with lead activated by electrolysis prior to use; D: using lead cathode and platinum anode.

b) Isolated yields after column chromatography on SiO_2 (hexane-AcOEt: 10/1~5/1).

carried out by using lead cathode and platinum anode without changing the current direction (Procedure D in the Table). In this case, 4-4.5 F/mol of electricity was required to complete the reduction and no detectable amount of lead(II) salts were recovered from the aqueous layer.

The results of the reduction of nitro alkenes 1 in Procedure A-D are summarized in the Table. The combination of aqueous perchloric acid and lead cathode is essential to each procedure, since the use of other materials as electrodes (cathode-anode: Pt-Pt, Cu-Pt, Cu-Cu, C-C, and stainless (SUS 27)-stainless) brought about no appreciable amount of reduction products 2 and 3 and the use of aqueous 10% sulfuric acid or aqueous 10% chloric acid resulted in the complete recovery of the starting materials.

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- 3) Ordinary, metal-acids reduction of 1 has been carried out at high temperature ($\sim 100^\circ\text{C}$) and the yields of 2 and/or 3 are insufficient (50-70% yields).
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- 5) The reported electrolytic reduction of nitro alkenes has been carried out in a divided cell under a strict control of the electrode potential and/or pH of the electrolysis medium.
- 6) Usual workup of the electrolyte afforded a mixture of 2a and 3a (59/33) in 92% yield.
- 7) J. Wiemann and O. Convert, C. R. Hebd. Seances Acad. Sci., Ser. C, 258, 4285 (1964).
- 8) Lead(II) salts dissolved in the aqueous layer was recovered as precipitates of lead(II) sulfate (226 mg).
- 9) Reduction of nitro alkenes with lead has not yet been reported.
- 10) Purchased from Kishida Chemical Co. Ltd., Osaka.

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